

A Computer-Vision-Based Bikeability Score

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Abstract—

Index Terms—Bikeability, computer vision, zero-shot classification, vision-language models, CLIP, surface classification, object detection, GPX.

I. INTRODUCTION

II. RESEARCH APPROACH AND METHODOLOGY

III. OBJECTIVES AND CONSTRAINTS

IV. GROUND DETECTION: ROAD-SURFACE CLASSIFICATION

The ridden surface is a primary determinant of cycling comfort and safety: a paved cycleway can be ridden effortlessly, whereas loose gravel or bare soil is strenuous and, on a road bike, potentially hazardous. Ground detection therefore assigns each sampled frame a single surface class and supplies the surface term of the bikeability score. Frames come from a handlebar-mounted camera at one frame per second; as each frame is dominated by one surface, the task is single-label image classification, not segmentation.

A. Task Definition and Model Selection

Two model families can solve this task without an existing labeled corpus: a classifier *trained on our own labels*, or a *zero-shot* vision-language model (VLM). We avoid a self-trained classifier, as it would need a large, balanced, manually labeled training set, careful regularization on limited data, and complete re-training whenever the class definitions change. A zero-shot VLM of the CLIP family [1] instead classifies an image from its similarity to natural-language class descriptions, so the taxonomy can be adapted by editing text – without task-specific training.

One distinction is often blurred: *linear probing* of self-supervised features (e.g., DINOv2 [2]) is *not* zero-shot – its linear head is fitted on labeled target data and is therefore supervised; it is excluded here and would only qualify as a declared supervised baseline.

B. Surface Classes and Prompts

We distinguish four cycling-relevant classes: **Cycleway** (a dedicated paved bike path), **Road** (a paved road for motor traffic), **Gravel** (loose crushed stone), and **Unpaved** (soil, mud, or field tracks). Each class is described by several English prompts whose normalized text embeddings are averaged (*prompt ensembling*). Since Cycleway and Road share the same smooth-asphalt texture, the prompts encode the *discriminating context* – a narrow separated lane with bicycle markings versus a wide multi-lane road – to push the two embeddings apart. A frame is assigned to the class of highest cosine similarity via a softmax.

C. Evaluation Protocol

We compare nine open CLIP-family models of light to medium size through the unified `open_clip` API [3]: OpenAI CLIP, OpenCLIP (LAION-2B, DataComp-XL) [4], SigLIP [5], MobileCLIP [6], and EVA-CLIP [7]. Both *accuracy* and *latency* matter for on-bike use. For the quantitative evaluation we manually labeled more than 1,700 frames from self-recorded rides; frames without a clear surface were tagged *unclear/other* and excluded. As the classes are imbalanced, we report accuracy, balanced accuracy and macro-F1 with 95% bootstrap confidence intervals and McNemar tests; latency is measured at batch size 1 with warm-up runs (ms/frame and FPS).

Table I
ZERO-SHOT SURFACE-CLASSIFICATION BENCHMARK ON THE SELF-LABELED TEST SET (> 1,700 FRAMES), SORTED BY MACRO-F1.
BACC: BALANCED ACCURACY; LATENCY AT BATCH SIZE 1.

Model	P[M]	Acc	Bacc	mF1	ms	FPS
OpenCLIP B/32 (LAION-2B)	151	0.75	0.72	0.58	115	8.7
SigLIP B/16 (WebLI)	203	0.75	0.62	0.57	259	3.9
CLIP B/16 (OpenAI)	150	0.78	0.62	0.57	261	3.8
MobileCLIP-S2	99	0.73	0.70	0.56	304	3.3
CLIP B/32 (OpenAI)	151	0.70	0.60	0.54	93	10.7
EVA02-B/16	150	0.76	0.62	0.54	304	3.3
OpenCLIP B/32 (DataComp-XL)	151	0.75	0.65	0.52	125	8.0
CLIP L/14 (OpenAI)	428	0.64	0.50	0.42	929	1.1

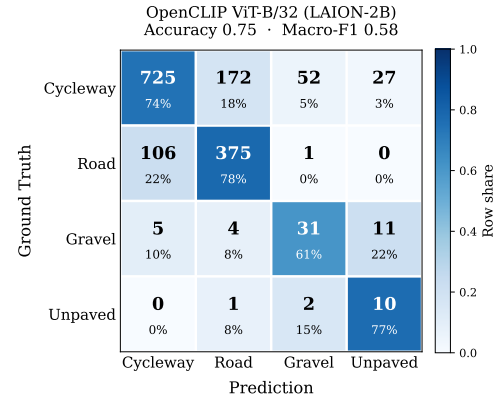


Figure 1. Row-normalized confusion matrix of the best model (OpenCLIP ViT-B/32, LAION-2B) on the labeled test set. Each cell shows the count and the row share; the diagonal is correct. Errors concentrate on the Cycleway/Road pair and the rare Gravel/Unpaved classes.

D. Results

Table I lists the outcome, sorted by macro-F1. The best model, **OpenCLIP ViT-B/32 (LAION-2B)**, reaches macro-F1 0.58 at only 115 ms/frame – the best accuracy-latency trade-off. Two points stand out. First, the highest plain *accuracy* (0.78, CLIP ViT-B/16) does *not* give the highest macro-F1: accuracy rewards the majority classes, whereas macro-F1 weights all four surfaces equally and is the more faithful measure on imbalanced data. Consistently, OpenCLIP ViT-B/32 (LAION-2B) also attains the clearly highest balanced accuracy (0.72 vs. ≤ 0.70 for every other model). Second, the 95% bootstrap accuracy intervals of the leading models overlap, so their differences are *not* statistically significant; only CLIP ViT-L/14 (428M) falls clearly below them – despite being the largest model it is both slowest and weakest (0.42 macro-F1), confirming that capacity alone does not help. As Fig. 1 shows, the residual errors concentrate on the Cycleway/Road pair (both smooth asphalt) and on the rare Gravel and Unpaved classes, which hold only 51 and 13 labeled frames and thus cap the macro-F1 at 0.58 – still sufficient for one of three sub-scores. We therefore adopt OpenCLIP ViT-B/32 (LAION-2B) as the ground detector. As all models share the same input and `open_clip` pipeline, the differences reflect architecture and training data; the CPU latencies are hardware-dependent.

V. OBJECT DETECTION

VI. ENVIRONMENT DETECTION

VII. COMPUTATION OF THE BIKEABILITY SCORE

The bikeability score combines the three sub-components – surface (ground detection), environment (environment detection), and objects (object detection) – into a single value in the interval $[0, 100]$. Every detector emits a fixed-length *vector*; each vector is first mapped to a *sub-score* in $[0, 1]$, and the three sub-scores are then combined by a weighted mean.

A. Detector Outputs as Vectors

The three detectors deliberately use different but fixed output representations, each aligned element-by-element with a stored value vector:

- **Ground** – a one-hot vector $\mathbf{g} \in \{0, 1\}^4$ over the classes (Cycleway, Road, Gravel, Unpaved); exactly one entry is 1.
- **Environment** – a vector of area fractions $\mathbf{f} \in [0, 1]^3$ over (vegetation, water, city).
- **Object** – a vector of counts $\mathbf{n} \in \mathbb{Z}_{\geq 0}^3$ over (bicycle, car, traffic light).

B. Overall Formula

For each evaluated frame, the score A results from the weighted mean of the three sub-scores:

$$A = 100 \cdot (w_g S_{\text{ground}} + w_e S_{\text{env}} + w_o S_{\text{object}}), \quad (1)$$

with the weights

$$w_g = 0.20, \quad w_e = 0.45, \quad w_o = 0.35, \quad w_g + w_e + w_o = 1. \quad (2)$$

Since all sub-scores lie in $[0, 1]$ and the weights sum to 1, the score is *guaranteed* to be bounded to $[0, 100]$.

C. Ground Sub-Score

The ground sub-score is the inner product of the one-hot vector $\mathbf{g}$ with the surface quality vector $\mathbf{q}^g = (1.0, 0.7, 0.3, 0.1)$ (Table II):

$$S_{\text{ground}} = \langle \mathbf{g}, \mathbf{q}^g \rangle = \sum_k g_k q_k^g. \quad (3)$$

Because $\mathbf{g}$ is one-hot, this simply selects the quality of the detected surface class.

D. Environment Sub-Score

The environment sub-score is the fraction-weighted mean of the environment quality vector $\mathbf{q}^e = (1.0, 1.0, 0.4)$:

$$S_{\text{env}} = \begin{cases} \frac{\langle \mathbf{f}, \mathbf{q}^e \rangle}{\sum_k f_k}, & \text{if } \sum_k f_k > 0, \\ 0.5, & \text{otherwise.} \end{cases} \quad (4)$$

The normalization by $\sum_k f_k$ keeps the sub-score in $[0, 1]$ even when the detected fractions do not cover the whole frame; a neutral value of 0.5 is used if no environment class is detected.

Table II

CLASS VALUE VECTORS FOR THE THREE SUB-SCORES: SURFACE QUALITY $\mathbf{q}^g$, ENVIRONMENT QUALITY $\mathbf{q}^e$, AND OBJECT PENALTY $\boldsymbol{\lambda}$ (HEURISTIC).

Component	Class	Value
Ground $\mathbf{q}^g$	Cycleway	1.0
	Road	0.7
	Gravel	0.3
	Unpaved	0.1
Environment $\mathbf{q}^e$	Vegetation	1.0
	Water	1.0
	City	0.4
Object penalty $\boldsymbol{\lambda}$	Bicycle	-0.2
	Car	+1.0
	Traffic light	+0.3

E. Object Sub-Score

The object sub-score maps the counts $\mathbf{n}$ through a *saturation function* using the penalty vector $\boldsymbol{\lambda} = (-0.2, 1.0, 0.3)$:

$$S_{\text{object}} = \exp\left(-\max(\rho, 0)\right), \quad \rho = \frac{1}{N} \langle \mathbf{n}, \boldsymbol{\lambda} \rangle, \quad (5)$$

with the number of aggregated frames N (here $N = 1$ per frame). A positive penalty for cars (+1.0) and traffic lights (+0.3) lowers the sub-score, while bicycle-friendly objects such as other cyclists carry a *negative* factor (-0.2) and raise it. The exponential form is robust against outliers, so a single crowded frame cannot dominate the overall score.

F. Route Processing over the Full Ride

The action camera splits one continuous recording into several files (e.g., DJI_0408, DJI_0409, ...). All parts are processed as a *single* continuous video: the files are ordered chronologically by the number in their name, and a cumulative time offset (the sum of the previous parts' durations) keeps the timeline gap-free across file boundaries. Within each part, one frame is sampled every 5 s by *seeking* directly to the target frame index instead of decoding every frame, which removes almost all of the decoding work. The three detectors are loaded once and evaluated in batches; for every sampled frame, the three output vectors and the resulting score are written to a shared CSV (frame name, continuous time in seconds and HH:MM:SS, an optional absolute ISO timestamp, the serialized vectors, and the score).

G. GPX Mapping and Visualization

Because the camera clock is unreliable, the frames are aligned to the GPS track by a single *manual sync point* (by default: recording start = GPX track start). Each frame is then matched to its nearest GPX track point in time (a nearest-neighbor join within a 3 s tolerance), which assigns a latitude/longitude to every score. The *average* bikeability score of the route is the mean over all M evaluated frames,

$$\bar{A} = \frac{1}{M} \sum_{i=1}^M A_i. \quad (6)$$

Finally, the route is drawn as a sequence of short segments, each colored by the local score on a red-orange-yellow-green

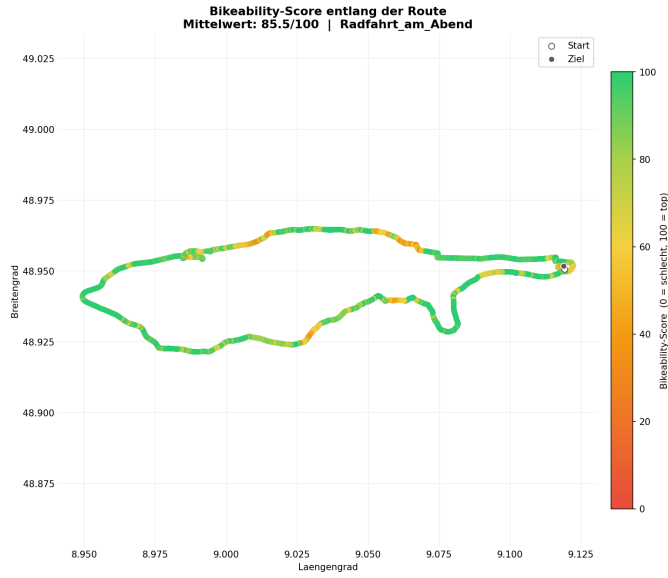


Figure 2. Bikeability score colored along the GPX route (red = low, green = high). Per-frame scores are matched to the nearest GPS track point via a manual sync point; the example ride has a mean score of $\bar{A} \approx 85.5/100$.

scale (Fig. 2). For the example ride shown, the mean score is $\bar{A} \approx 85.5/100$; the track is predominantly green (good cycling conditions) with a few yellow/orange sections.

VIII. DISCUSSION

IX. CONCLUSION

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